EPSR-test-ex66 lsi: water local structure index, 3 chained runs

Program(s): lsi | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises `lsi`, which computes a local structure index: for each molecule, bonded and non-bonded near neighbours are found with the bond criteria of `chains`, and the LSI is the distance of the nearest non-bonded minus that of the furthest bonded neighbour (method of Shi, Russo and Tanaka, J. Chem. Phys. 149, 224502 (2018), as cited in `lsi.f`; the LSI concept goes back to Shiratani and Sasai [1]). Neighbour counts, neighbour-triangle angles and r3..r6 distributions are also written.

Code exercised

Fortran (original, fortran_src/):

- `lsi.f`: `run_lsi` (running averages `nbonded`, `nnonbonded`, `avbond`), `find_lsi` (bonded / non-bonded neighbours, LSI, triangles, r3..r6)
- `chains_routines.f`: `match`
- `read_and_save_files.f`: `outgofz`, `save_files_others`; `setup_symbols_generic.f`: `setup_symbols_lsi`

Rust (port, rust/epsr-core/src/):

- `tools/lsi.rs`: `run`, `find_lsi`, `dfs_range` (out-of-bounds emulation)
- `tools/chains.rs`: `read_criteria`, `fmatch`, `set_nsize_default`; `tools/clusters.rs`: `outgofz`
- `symbols.rs`: `setup_input_file`, `read_input_file`, `save_inp_file`; `output.rs`: `r_int`; `symtab_tools.rs`: `LSI`

Physical system

Analysed configuration: `200DMSO400Water.ato`, an unchanged copy of the EPSR-test box of 200 dimethyl sulphoxide (DMSO; atom types S, O, C, H) and 400 water molecules (O1, H1): 3200 atoms, DMSO mole fraction 1/3, 300 K, cubic box of side 33.1213 Å (0.0881 atoms/Å^3), the EPSR model of the deuterated DMSO-water neutron data (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), cf. [2]. Water O1-H1···O1 hydrogen-bond criterion (O-H 0.8-1.2, O···O 2.4-3.4, H···O 1.4-2.5 Å, angle <= 30°), LSI range -1 to 2 Å in 100 bins. After three passes the reference gives 1.40 bonded and 0.34 non-bonded neighbours per O1 and a mean bonded O···O distance of 2.80 Å.

Inputs

`water.LSI.dat`: `nsize 100`, `lsiminmax -1.0 2.0`, `nsumt 0`, one criterion. `200DMSO400Water.ato` copied unchanged from EPSR-test; the keyword file was written by hand (the Fortran program creates no template when the named file does not exist; keywords and defaults from `setup_symbols_generic.f`).

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
`repeat 3`. Runs 1 and 2 use fc = 1 (the Fortran uses the input nsumt: 1/nsumt only for nsumt > 0), run 3 fc = 0.5 for the running averages written back to the `.dat` file (format `2(1x,e12.5)`, second value the spread). Compared after each pass: `water.LSI.n01` (10 columns: bonded/non-bonded counts, LSI, three triangle-angle distributions, r3..r6) and `water.LSI.dat`. The configuration is the same in every pass, so the chained passes test the read-back of the accumulated file and the running-average arithmetic.

Notes

DMSO molecules have no O1, so their neighbours have no matching atom pair (`iwmin` = `ia2min` = `0`); the Fortran then reads `dfs1(id,0)` and `xa(0)` out of bounds; this is emulated in the port.

References

[1] E. Shiratani and M. Sasai, "Growth and collapse of structural patterns in the hydrogen bond network in liquid water", J. Chem. Phys. 104, 7671-7680 (1996).
[2] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[3] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.